

MIRAGE-BENCH: Automatic Multilingual Benchmark Arena for Retrieval-Augmented Generation Systems

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Abstract

Traditional retrieval-augmented generation (RAG) benchmarks evaluate systems using heuristic-based metrics, but these require human preferences as the ground truth for reference. In contrast, arena-based benchmarks, where systems compete against each other, require an expensive large language model (LLM) as a judge for a reliable evaluation. We present a simple efficient technique to combine the best of both worlds. The idea is to train a surrogate judge using heuristic metrics as input, to output the LLM as a judge prediction. In our work, we develop MIRAGE-BENCH,¹ a synthetic arena-based RAG benchmark for 18 diverse languages on Wikipedia focused on multilingual answer generation evaluation. It extensively couples both heuristic features and LLM as a judge for evaluation. We benchmark 19 multilingual LLMs, and observe a high correlation (Kendall Tau (τ) = 0.909) using our surrogate judge and between GPT-4o as a teacher using the Bradley-Terry framework. Our results show proprietary and large open-source LLMs currently dominate on MIRAGE-BENCH. Our code and datasets are made publicly available here: <https://github.com/vectara/mirage-bench>.

1 Introduction

Large language models (LLMs) have recently gained popularity for information-seeking queries leading to the widespread adoption of retrieval-augmented generation (RAG) (Guu et al., 2020; Lewis et al., 2020; Izacard and Grave, 2021; Borgeaud et al., 2022). The naive RAG setup traditionally includes a retrieval and a generation stage, conducted sequentially. RAG systems such as Bing Search (Microsoft, 2023) provide grounded responses, i.e., statements that include citations to one or more retrieved passages. The citations

¹MIRAGE-BENCH has been coined as (MIRACL + RAG + EVALUATION + BENCHMARK).

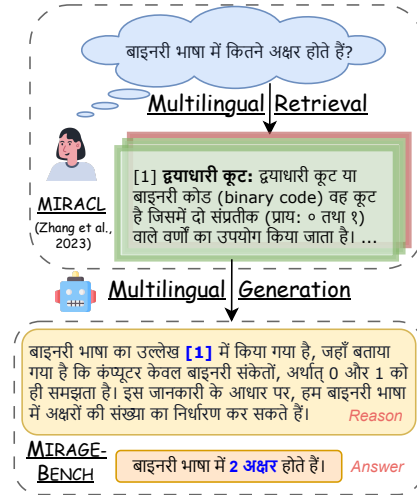


Figure 1: Multilingual naive RAG pipeline in Hindi (hn). In MIRAGE-BENCH, we reuse the oracle retrieval set (query and oracle judged passages) from MIRACL (Zhang et al., 2023) and focus on evaluating the answer generation stage with multilingual LLMs.

reduce factual hallucinations with easy verifiability and improve faithfulness to passages provided within context (Khandelwal et al., 2020; Lewis et al., 2020; Gao et al., 2023a,b; Liu et al., 2024). However, existing RAG benchmarks are English-centric, due to uneven and scarce data available across multiple languages (Thakur et al., 2024c). So far, it is unclear how existing LLMs perform in multilingual RAG, i.e., where queries and passages are non-English and the LLM generates the answer in the same language. An example of a naive RAG pipeline in Hindi (hn) is shown in Figure 1.

RAG benchmarks can be broadly classified as either (i) *heuristic-based*, where benchmarks handcraft evaluation metrics (e.g., faithfulness or fluency) to evaluate systems on multiple dimensions (Gao et al., 2023a; Chen et al., 2024c; Yang et al., 2024b) or (ii) *arena-based*: where systems compete each other in a tournament and an LLM-based teacher is used as the judge for evaluation

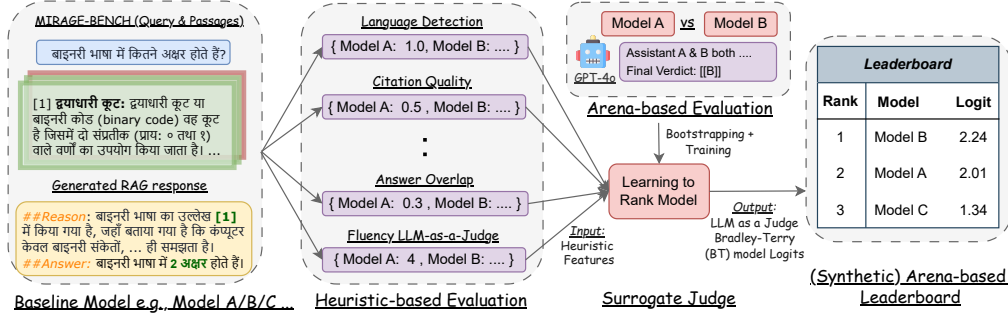


Figure 2: The MIRAGE-BENCH evaluation flowchart consists of three steps: (i) heuristic-based features evaluating the baseline model response across several dimensions; (ii) exhaustive pairwise comparisons with GPT-4o as a judge on a small subset of queries to train our surrogate judge. (iii) After training, we utilize our surrogate judge to output the model ranking on the whole subset of queries, to construct the synthetic RAG arena-based leaderboard.

(Rackauckas et al., 2024; Pradeep et al., 2024a). Heuristic-based benchmarks are computationally cheaper to evaluate but require human preferences as gold truth for reference. They also face challenges in aggregating different features into a ranking order for models. On the other hand, arena-based benchmarks require a high-performance or reasoning-intensive LLM as a teacher for accuracy (Zheng et al., 2023), which makes exhaustive pairwise comparisons expensive for a large set of models. For instance, evaluating a single query on 19 models involves $\binom{19}{2} = 171$ comparisons and costs between \$5–10 USD with GPT-4o (OpenAI, 2023).

In our work, we get the best of both worlds with a surrogate judge, a learning to rank model, e.g. random forest (Ho, 1995), using heuristic features to estimate an arena-based leaderboard obtained with a Bradley-Terry model (Hunter, 2004) from pairwise evaluations using an LLM as a judge (Zheng et al., 2023). We use bootstrapping to obtain confidence bounds for better statistical estimates. After training, the learning to rank model can be used to estimate the performance of newer released models reliably in the future without the expensive LLM as a judge. It also provides better interpretability and is easily retrainable with a different or newer set of heuristic features.

We develop MIRAGE-BENCH, a RAG benchmark across 18 languages for multilingual generation evaluation on Wikipedia. MIRAGE-BENCH data is sourced from MIRACL (Zhang et al., 2023), a multilingual retrieval dataset containing human-generated queries and human-labeled relevance judgments for Wikipedia passages. We benchmark 19 frontier multilingual LLMs in our experiments. Our evaluation flowchart adopted in

MIRAGE-BENCH is shown in Figure 2. We evaluate seven heuristic features: (i) language detection, (ii) citation quality, (iii) support, (iv) reranker score, (v) answer overlap (traditional), (vi) answer overlap (LLM-measured), and (vii) fluency (LLM-measured). We use GPT-4o as the judge to evaluate our pairwise RAG comparisons on a smaller subset consisting of 100 queries. Next, we train a random forest model as a surrogate judge, using the heuristic features as input, and learn to output the Bradley-Terry model logits as output. Finally, during inference, we use our surrogate judge to produce a “synthetic” ranking leaderboard for all baselines across every language.

More specifically, we ask the following research questions in our work:

- Can we avoid the LLM as a judge cost and combine heuristic-based evaluation?
- How do frontier multilingual-focused LLMs perform in multilingual answer generation?
- Does fine-tuning on MIRAGE-BENCH training dataset improve LLM performance?

Our experimental results show that: (i) random forest model (surrogate judge) learned rankings highly correlates with GPT-4o as a judge, achieving an average Kendall-Tau (τ) score of 0.909. (ii) proprietary and large open-source LLMs (≥ 70 B parameters) achieve the topmost ranks in the MIRAGE-BENCH leaderboard. (iii) MIRAGE-BENCH training data, synthetically constructed using GPT-4o, is beneficial to improve smaller open-source LLMs (7–8B parameters). The main contribution of work is building MIRAGE-BENCH and benchmark 19 frontier multilingual LLMs to accelerate development in the area of multilingual RAG.

	ar	bn	de	en	es	fa	fi	fr	hi	id	ja	ko	ru	sw	te	th	yo	zh
MIRAGE-BENCH Evaluation Dataset																		
# Queries	1501	411	304	787	617	632	1183	343	350	939	797	213	1247	481	150	730	119	391
# Avg. Query Tokens	10.2	17.6	10.4	9.1	11.4	14.1	12.2	10.4	17.3	10.1	14.6	14.2	14.3	12.0	16.1	19.6	13.0	8.0
Avg. Rel. Passages / Q	2.0	2.1	2.6	2.8	4.3	2.1	2.0	2.1	2.1	3.1	2.2	2.6	2.8	1.9	1.2	1.8	1.2	2.5
Avg. Non Rel. Passages / Q	8.1	8.0	7.7	7.7	5.6	8.3	8.1	7.9	7.8	7.0	8.2	11.8	7.6	8.7	4.9	8.5	8.8	7.5
MIRAGE-BENCH Training Dataset																		
# Queries	3468	1624	—	2857	2159	2104	2878	1137	1165	4054	3466	859	4567	1866	3283	2965	—	1311
# Avg. Query Tokens	10.2	17.4	—	8.7	11.2	14.2	11.8	10.4	17.1	9.9	14.5	14.9	14.3	12.0	16.1	19.3	—	7.9
Avg. Rel. Passages / Q	1.8	2.3	—	2.5	3.6	2.0	1.7	2.0	2.0	2.9	1.9	2.1	2.0	1.4	1.2	1.6	—	2.3
Avg. Non Rel. Passages / Q	5.5	7.9	—	7.5	5.3	8.3	5.3	8.0	7.9	7.1	7.9	12.4	5.2	3.5	4.3	5.6	—	7.6
# Avg. GPT-4o Context Tokens	105.5	144.5	—	137.9	203.4	133.8	129.4	180.1	149.4	100.8	140.2	136.4	90.0	106.4	54.6	86.0	—	137.9
# Avg. GPT-4o Answer Tokens	35.7	22.9	—	20.6	55.3	51.6	30.3	56.0	26.6	22.6	24.1	23.9	16.1	18.4	11.4	20.4	—	40.6

Table 1: Dataset statistics for 18 languages in MIRAGE-BENCH; All tokens are calculated using the GPT-4o tokenizer (OpenAI, 2023); (Rel.) denotes relevancy; (# Avg GPT-4o) counts the tokens in the GPT-4o generated context and answer; Training data is not available for two languages (denoted by —): German (de) and Yoruba (yo).

2 Related Work

Prior work on RAG evaluation has been conducted exclusively in English. For example, benchmarks such as ALCE (Gao et al., 2023a), FreshLLM (Vu et al., 2024), ClapNQ (Rosenthal et al., 2025), HAGRID (Kamalloo et al., 2023) and CRAG (Yang et al., 2024b), all include long-form answers for English-only queries and are based on collections containing documents from either the English Wikipedia, MS MARCO (Bajaj et al., 2016) or NQ (Kwiatkowski et al., 2019). Similarly, TREC 2024 RAG, a TREC competition for RAG evaluation is focused on evaluating queries in English.²

Multilingual RAG. On the multilingual side, RAG has not been well studied in prior literature. The RGB benchmark (Chen et al., 2024c) is limited in language scope covering only one additional language: Chinese (zh). NeuCLIR 2024 track (Mayfield et al., 2024) evaluates long-form report generation from participants, but is limited to three languages. A concurrent work is BERGEN (Chirkova et al., 2024), which evaluates the multilingual open-domain QA setting on 13 languages. In contrast, in MIRAGE-BENCH, we (i) evaluate the generation task in the multilingual RAG pipeline on 18 languages, (ii) provide multilingual instruction-tuned data for RAG fine-tuning, and (iii) utilize high-quality native-speaker multilingual queries generated in MIRACL (Zhang et al., 2023).

Learning to rank. A supervised learning technique, where models are trained to rank items within a list similar to training data (Liu, 2010; Turnbull, 2017). Models are trained in either a pointwise, pairwise, or listwise objective (Cao et al., 2007). In our work, we experiment with simple models such as random forest to train to

rank the Bradley-Terry model coefficient produced by LLM as a judge. We keep complex models such as LambdaMART (Burgess, 2010) for future work.

3 An Overview of MIRAGE-BENCH

We select 18 languages in MIRAGE-BENCH as the starting point, representing an appropriate cross-section of the diversity of the languages spoken worldwide at this point. MIRAGE-BENCH is a comprehensive multilingual RAG benchmark focusing on the generation task evaluation. As shown in Table 1, the benchmark includes 11,195 evaluation pairs and 39,763 training pairs across 18 languages.

3.1 Distinction and Extension from MIRACL

MIRACL introduced in Zhang et al. (2023) is a monolingual retrieval dataset, which evaluates the retrieval task, i.e., given a user query and a passage corpus, retrieve the ranked list of passages as output. MIRACL contains human-annotated relevance judgments to evaluate retrieval and re-ranker models, e.g., lexical models like BM25 (Robertson and Zaragoza, 2009) or bi-encoders like mDPR (Karpukhin et al., 2020), or late-interaction models like ColBERT (Khattab and Zaharia, 2020).

In contrast, MIRAGE-BENCH evaluates the generation task in RAG, requiring LLMs to generate a summarized answer given the query and context available from oracle-judged passages. In our work, we reuse the queries and relevance-judgments from MIRACL, but solely evaluate the multilingual generation task in RAG measuring answers using both heuristics and LLM as a judge. Since MIRAGE-BENCH evaluates the generation task containing the oracle context, independent of the retrieval task, the chances of contamination in MIRAGE-BENCH evaluation from fine-tuning MIRACL is less overall.

²TREC 2024 RAG: <https://trec-rag.github.io/>

3.2 MIRAGE-BENCH Evaluation Dataset

MIRACL queries are high-quality and generated by native speakers (Zhang et al., 2023). The annotation procedure in MIRACL is identical to TyDI-QA (Clark et al., 2020). The passage collection is constructed from language-specific Wikipedia corpora and parsed using WikiExtractor. The MIRAGE-BENCH evaluation dataset is constructed re-using the queries and oracle-judged passages available in the MIRACL development split.³

We incorporate two changes: (i) In Arabic (ar), we randomly sample a smaller subset of 1,501 out of 2,896 queries for uniformity in the number of queries available across other languages. (ii) We filter out a small subset of queries with zero non-relevant passages, i.e., we always include queries with hard passages from MIRACL to make the MIRAGE-BENCH evaluation challenging.⁴

3.3 MIRAGE-BENCH Training Dataset

The MIRAGE-BENCH training dataset is developed from the MIRACL training dataset (Zhang et al., 2023). In MIRAGE-BENCH, we reuse all the MIRACL training pairs available in 16 languages (except German (de) and Yoruba (yo)) and convert it using a simple recipe into a multilingual instruction-tuned RAG dataset (Zhang et al., 2024; Niu et al., 2024). In our recipe, we first keep only the relevant passages as context along with the input query to generate a zero-shot RAG output by prompting strong teachers such as GPT-4o (OpenAI, 2023), Llama 3 (70B) (Dubey et al., 2024) and Mixtral (8×22B) (Jiang et al., 2024). After generation, we include non-relevant passages within our prompt as “distracting and noisy context”, to help improve the quality of the training dataset. Since we convert a retrieval dataset, we do not have human-annotated answers for questions in MIRAGE-BENCH.

4 Multilingual RAG Evaluation

4.1 Heuristic-based Evaluation

Answer generation in RAG requires evaluation on various dimensions. For example, whether a system’s response provides the correct final answer or cites the relevant documents, a single metric alone is *not sufficient* to capture the comprehensive eval-

uation required for RAG systems. Inspired by other recent works (Kiela et al., 2021; Santhanam et al., 2023; Gao et al., 2023a), we introduce five *deterministic* features and two *LLM-measured* features for evaluation in our work. We rely on features that are explainable, cheap, and fast to compute. We explain each heuristic feature in Appendix A.

Language detection. We compute the probability of a system’s response in the required target language with langid (Lui and Baldwin, 2012). We compute two metrics: language detection (target language) and English detection.

Citation quality. Using passage-level relevance judgments for all queries (or qrels) information available in MIRACL, we evaluate whether the system’s response cites the relevant passages, crucial for measuring faithfulness. We compute and evaluate: Recall@ k and MAP@ k , where $k = 10$, as we have a maximum of 10 passages per query.

Support. Grounding is necessary to avoid hallucinations in the system’s response. Support evaluation (Gao et al., 2023a) checks whether each sentence is supported by cited passages using a multilingual NLI model (He et al., 2023).⁵ We compute the probability of the *entailment* and *neutral* score, macro-averaged across the sentence-citation pairs.

Reranker score. The reranker score measures the average similarity (can be greater than 1.0) between the query and the passages cited within the system’s response. We compute the reranker score using a multilingual reranker model,⁶ macro-averaged across the query-passage pairs.

Answer overlap. Having the correct answer is crucial in the RAG system’s response. Since MIRAGE-BENCH does not include a human-labeled answer, we use the generated answer from GPT-4 (OpenAI, 2023) as the gold truth. We compute two traditional metrics: SacreBLEU (Papineni et al., 2002) and ROUGE-L (Lin, 2004) measuring the lexical word overlap between the gold answer (GPT-4’s answer is used for reference) and the system’s response.

Answer overlap (LLM-measured). In addition, we evaluate using Llama-3 (8B) (Dubey et al., 2024), an open-source LLM as a judge evaluator in a pointwise setup, providing a semantic word overlap integer score in the range [1, 5]. The answer overlap prompt description is listed in Figure 11.

³We did not utilize the test split in MIRACL as the relevance judgments are not publicly available.

⁴An exception is Telugu (te), where 78 queries have at least one non-relevant passage. Therefore, we randomly sample 72 additional queries with only relevant judged passages.

⁵MoritzLaurer/mDeBERTa-v3-base-xnli-multilingual-nli-2mil7

⁶Reranker (Chen et al., 2024b): BAAI/bge-reranker-v2-m3

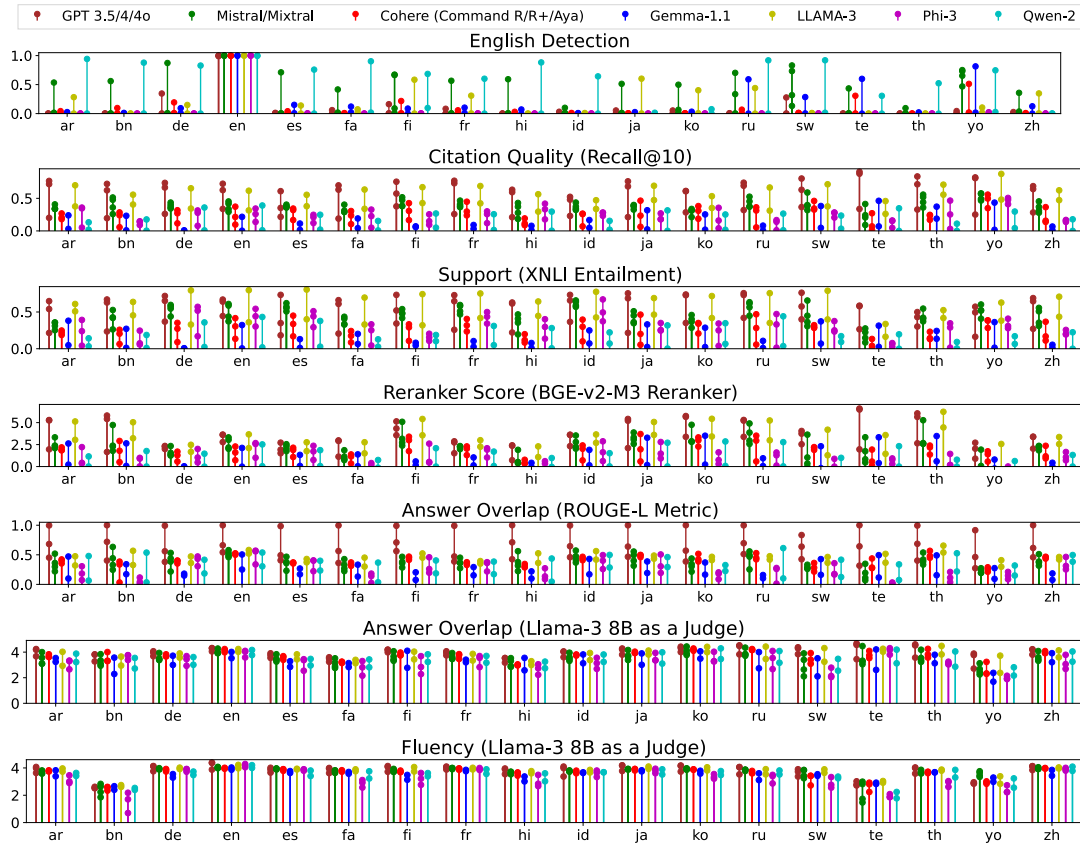


Figure 3: Lollipop plots denoting the average heuristic-based feature scores achieved by LLM baselines for each language in MIRAGE-BENCH. x -axis denotes the 18 languages; whereas y -axis plots every heuristic feature score. Models in the same LLM family are represented in the same color in a lollipop (as multiple circles). Figure 9 in the Appendix provides lollipop plots for all eleven heuristic-based features used in our work.

Fluency (LLM-measured). It measures for grammatical correctness and idiomatic word choices in the system’s response. As previously mentioned, we use Llama-3 (8B) (Dubey et al., 2024) in a pointwise setup, proving an integer score in $[1, 5]$. The fluency prompt description is listed in Figure 12.

4.2 Arena-based Evaluation

Heuristic-based evaluation metrics often rely on a gold standard for evaluation. Tasks such as text retrieval (Bajaj et al., 2016; Thakur et al., 2021) require human-labeled relevance judgments, and similarly, NLP tasks such as machine translation (Stahlberg, 2020), require human-annotated translations. As human preferences are seldom available in numerous applications, using LLM as a judge (Zheng et al., 2023; Chen et al., 2024a; Chiang et al., 2024) is becoming a de facto approach for arena-based evaluation of LLMs.

We evaluate pointwise, listwise, and pairwise

LLM as judge evaluations. We anecdotally observe the pointwise judge, which is efficient $O(n)$, but is not good for ranking LLMs as it provides similar scores for a wide range of models (e.g., 4/5 score for 16 out of 18 LLMs evaluated). Similarly, the listwise judge finds it difficult to rank all 19 models in the correct order. Therefore, although suboptimal in complexity, $O(n^2)$, where n is the number of models evaluated, we choose a pairwise evaluation in our work.

Pairwise LLM as a judge. Following prior works on arena-based evaluation in RAG (Rackauckas et al., 2024; Pradeep et al., 2024a), we evaluate two system’s responses in a head-on comparison by computing pairwise judgments with LLM as a judge. We reuse the RAGelo prompt template (Rackauckas et al., 2024) with minor additions. The prompt template is listed in Figure 13. LLM as a judge evaluator includes three types of biases (Zheng et al., 2023): (i) verbosity bias (Wu and

Aji, 2025) (ii) self-enhancement bias (Xu et al., 2024; Panickssery et al., 2024) and (iii) position bias (Wang et al., 2024). We avoid the verbosity bias, as RAG evaluation has fixed evaluation criteria requiring sentence-level citations and answers (Pradeep et al., 2024a) and the position bias by randomly swapping the position of two models.

4.3 Learning to Approximate Rankings

There is no predefined way to aggregate the heuristic features to provide an overall leaderboard ranking in MIRAGE-BENCH. Averaging the scores is too simplistic as features measure different aspects of RAG evaluation. On the other hand, arena-based evaluations provide ranked leaderboards but are computationally expensive to compute with a strong teacher model. To avoid computational costs, smaller models as teachers have been proposed (Thakur et al., 2024a; Ni et al., 2024).

Motivated by similar observations, we train a surrogate judge to effectively emulate an arena-based leaderboard without incurring the expensive LLM as a judge pairwise cost. We find the random forest model to serve as a scalable and cost-effective judge that can be trained within minutes on small training datasets without expensive computation. Therefore, we train a random forest (learning to rank model) as a surrogate judge to approximate the Bradley-Terry model coefficients (Hunter, 2004) learned from an arena-based evaluation that uses GPT-4o as a judge for pairwise judgments.

Learning to rank model. While the heuristic features introduced in Section 4.1 can be computed efficiently and without the reliance on proprietary LLMs, inducing a ranking from pairwise comparisons via a Bradley-Terry model is computationally expensive and requires access to a high-performance LLM. Furthermore, as we demonstrate in Section 6.3, the ranking accuracy, measured by the Kendall-Tau (τ) coefficient, degrades rapidly when subsampling tournament matches.

The procedure, detailed in Algorithm 1, simulates N_t tournaments, each involving a total of N_l models and N_q queries. For each query, judgments are obtained for all $\binom{N_l}{2}$ pairings of models. We employ bootstrapping on the query selection process to estimate the variance in the R^2 metric in the learning to rank models’ approximations of the Bradley-Terry coefficients over a randomly-sampled holdout set, $LLM_{predict}$.

We randomly select two models, Gemma 1.1

(2B) and Llama-3 (70B) as holdout models, i.e., we do not train on the features for holdout models. For English, we observe an average $\bar{R}^2 = 0.971$ with a 95% confidence interval of $[0.905, 0.999]$. On the other hand, for Bengali, we observe $\bar{R}^2 = 0.937$ with a 95% confidence interval of $[0.766, 0.998]$. $\bar{R}^2$ scores for all 18 languages with 95% confidence intervals are listed in Table 5. Taken together, these results indicate that the training procedure is fairly robust with $N_q = 100$.

Algorithm 1 Simulate Tournaments and Fit Models

```

1: for  $i \in [N_t]$  do
2:    $M_{BT}^i \leftarrow \text{TOURNAMENT}(N_q)$ 
3:    $X_t, Y_t \leftarrow \text{DATASET}(LLM_{train}, M_{BT}^i)$ 
4:    $X_p, Y_p \leftarrow \text{DATASET}(LLM_{predict}, M_{BT}^i)$ 
5:    $M_{reg}^i \leftarrow \text{FIT}(X_t, Y_t)$ 
6:    $R_2^i \leftarrow M_{reg}^i(X_p, Y_p)$ 
7: end for
8:  $M_{BT} \leftarrow [M_{BT}^1; M_{BT}^2; \dots; M_{BT}^{N_t}]$ 
9:  $M_{reg} \leftarrow [M_{reg}^1; M_{reg}^2; \dots; M_{reg}^{N_t}]$ 
10:  $R_2 \leftarrow [R_2^1; R_2^2; \dots; R_2^{N_t}]$ 

```

Note: Refer to Section 4.3 for a definition of each of the variables. The TOURNAMENT function runs a battle arena, sampling q queries, and returning the learned Bradley-Terry model. The DATASET function accepts a set of LLMs and a learned Bradley-Terry model. It returns X , the heuristic RAG feature values, and Y , the Bradley-Terry coefficients for each LLM model. After simulating N_t tournaments, the array R_2 contains the R^2 errors for each of the N_t models.

5 Experimental Settings

5.1 Multilingual Baselines

Existing frontier LLMs are either English-only or support a limited set of languages, predominantly due to the *curse of multilinguality* for large models (Conneau et al., 2020). It is unclear how well existing LLMs perform on RAG on a wide variety of languages, due to the scarce availability of multilingual instruction tuning datasets. We experiment with LLMs from seven different families, containing proprietary and open-source LLMs. Wherever possible, we evaluate the *instruction-tuned* version if available. Refer to Appendix B for more details.

- **OpenAI:** GPT-3.5-turbo, GPT-4, and GPT-4o (OpenAI, 2023) using the Azure OpenAI service.
- **Mistral:** Mistral-Instruct-v0.2 and v0.3 (7B) (Jiang et al., 2023) and Mixtral-Instruct-v0.1 (8×7B) and (8×22B) versions (Jiang et al., 2024).
- **Cohere:** Command-R (35B), Command-R+

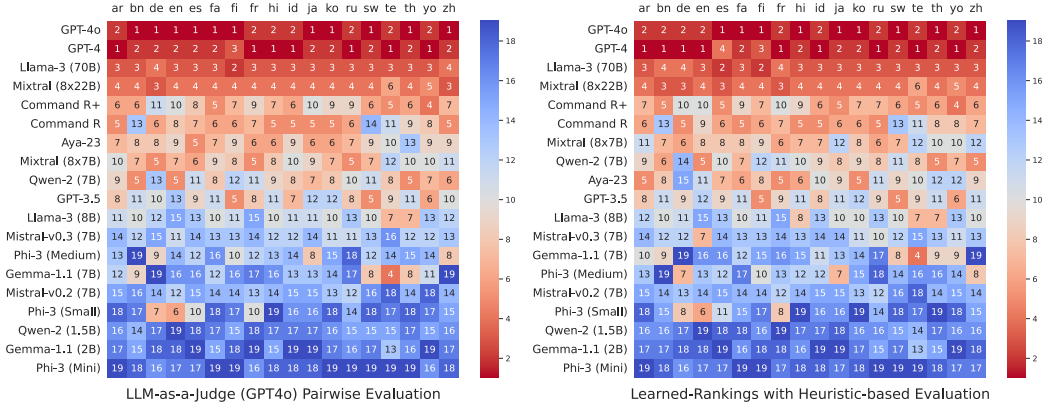


Figure 4: MIRAGE-BENCH arena-based leaderboards: (left heatmap) Bradley-Terry model coefficients with GPT-4o as a pairwise judge for a subset of 100 sampled queries; (right heatmap) Synthetic rankings using heuristic-based features and a random forest model as a surrogate judge on all queries. Each highlighted cell denotes the rank of the LLM (lower the better). LLMs are sorted by lowest to highest average rank across all 18 languages.

- (104B) and Aya-23 (35B) (Aryabumi et al., 2024).
- **Gemma:** Gemma 1.1 instruct (2B) and (7B) models (Mesnard et al., 2024).
- **Llama-3:** Llama-3 instruct (8B) and (70B) models (Dubey et al., 2024).
- **Phi-3:** Phi 3 instruct series: Medium (14B), Small (7B), and Mini (3.8B) (Abdin et al., 2024).
- **Qwen-2:** Qwen-2-instruct series: 1.5B and 7B (Yang et al., 2024a).

Prompt template. We internally optimized⁷ the ChatQA prompt template (Liu et al., 2024), to include in-text citations of the context passages following the IEEE format (Kamalloo et al., 2023). In MIRAGE-BENCH, we have about 10 passages annotated in the oracle setting. Therefore, we trim each passage available and take the first 800 tokens to fit all passages within a fixed context length of 8192 tokens. following prior work in Shi et al. (2023), the prompt requires the LLM to explain the multilingual generation task starting with “##Reason” and the answer itself starting with “##Answer”. Utilizing this output format has its advantages in easily parsing the generated answer and the rationale behind the answer. The prompt template for multilingual generation is shown in Figure 10.

6 Experimental Results

6.1 Heuristic-based Results

Figure 3 shows lollipop plots indicating the average heuristic-feature value (y -axis) distribution for each

⁷A majority of the prompt optimization was internal and based on eye-balling RAG responses across LLMs.

Lang.	τ	Lang.	τ	Lang.	τ	Lang.	τ
ar	0.951	bn	0.874	de	0.825	en	0.835
es	0.876	fa	0.924	fi	0.949	fr	0.914
hi	0.946	id	0.896	ja	0.892	ko	0.950
ru	0.849	sw	0.958	te	0.938	th	0.946
		yo	0.906	zh	0.941		
Avg. Kendall Tau (τ) on 18 languages = 0.909							

Table 2: Kendall τ rank correlation between pairwise GPT-4o as a judge Bradley-Terry model and the synthetic arena-based ranking leaderboard generated using our surrogate judge in MIRAGE-BENCH.

language (x -axis). In English detection (higher the worse), smaller LLMs such as Gemma-1.1 (2B) do not generate output in the required target language but rather generate reasoning and answers in English. For citation quality, support, and reranker score features, LLMs from OpenAI and Llama-3 family achieve high Recall@10 and entailment scores (except Llama-3 (70B) for a few languages), indicating generated answers include grounded citations from relevant passages. In contrast, LLMs from the Qwen-2 or Gemma-1.1 family, tend to under-cite passages in their answers.

Furthermore, we observe LLMs from the OpenAI family achieve the highest word overlap in the ROUGE-L metric (GPT-4 used as ground truth, which could be a potential bias) and Llama-3 (8B) as a judge, while we observe less variance across other LLMs. In Fluency, we observe a majority of the LLMs are rather fluent in generation, except Bengali (bn), Telugu (te), and Yoruba (yo).

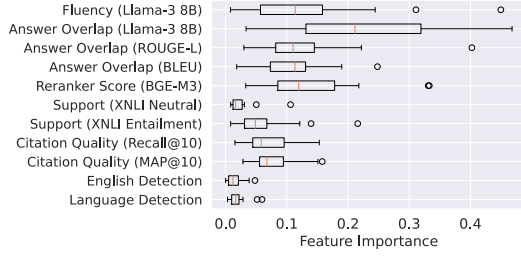


Figure 5: Boxplot with the feature importance value (averaged across 18 languages in MIRAGE-BENCH) observed by the learning to rank (random forest) model.

6.2 Arena-based Results

Figure 4 (left heatmap) is the arena-based leaderboard with the Bradley Terry model conducting 200 tournaments and bootstrapping 100 matches per tournament on a subset of 100 queries using GPT-4o as a pairwise judge. We observe that proprietary LLMs such as GPT-4o and GPT-4, and larger open-source LLMs such as Llama-3 (70B) and Mixtral (8×22B) perform better than other LLMs. LLM rankings across languages are usually stable; with a few notable exceptions such as Gemma-1.1 (7B) which achieves a rank of 4 in Telugu. Command R (35B) performs poorly in low-resource languages such as Bengali (rank 13) or Swahili (rank 14). The complete scores including model coefficient logits and 95% confidence intervals (error bars) are provided in Table 7 and Table 8 in the Appendix.

Synthetic rankings using random forest. Figure 4 (right heatmap) is the learned synthetic leaderboard rankings on all queries using heuristic-based features trained with a random forest model. The synthetic leaderboard generated using a surrogate judge, highly correlates to the GPT-4o as a pairwise judge, achieving an average Kendall-Tau (τ) rank correlation = **0.909**, by training on 17 LLMs during training and keeping 2 LLMs as a holdout for every language. Individual language-specific Kendall-Tau rank correlation scores are listed in Table 2. This provides evidence of the efficacy of training a random forest model as a surrogate judge. In Appendix D, we extend the evaluation for Llama-3.1 and Gemma-2 series of LLMs.

Heuristic feature importance. In Figure 5, we plot the average feature importance achieved by our random forest model as a surrogate judge. Using Llama-3 (8B) as a judge, for fluency and answer overlap are the important heuristic features. Similarly, deterministic answer overlap and reranking-

Model / Language	ar	bn	fi	ja	ko	ru	te
Train R^2 on randomly selected fifteen models							
Random Forest	0.97	0.96	0.97	0.96	0.97	0.95	0.97
Linear Regression	0.98	0.98	0.98	1.00	0.97	0.99	1.00
MLP Regressor	0.97	0.98	0.97	0.96	0.96	0.98	0.99
XGB Regressor	1.00	1.00	1.00	1.00	1.00	1.00	1.00
SVR	0.75	0.77	0.81	0.59	0.67	0.73	0.39
Holdout R^2 on four randomly selected held out models							
Random Forest	0.50	0.41	0.49	-0.03	0.45	0.07	0.83
Linear Regression	-1.92	-5.45	-19.38	-0.06	-26.53	-2.13	0.31
MLP Regressor	0.33	0.37	0.45	-0.76	-0.04	-0.48	0.78
XGB Regressor	-0.02	0.44	0.22	-1.33	-0.09	-0.80	0.59
SVR	-0.03	0.48	0.64	-0.53	-0.09	0.15	-0.10

Table 3: Train and Holdout R^2 scores using different learning to rank model choices. Each experiment has been repeated 50 times with four held-out models.

based metrics are equally important. Some heuristic features such as language detection, and neutral score in support evaluation obtain the least importance. We observe all answer-related heuristic features achieve a high importance indicating that the generated “answer” portion in an LLM’s response is crucial and required to learn rankings from GPT-4o as a pairwise judge.

6.3 Ablations & Discussion

To better understand the gaps observed during training of the random forest model as a surrogate judge, we conduct further ablations on a subset of seven languages including Arabic (ar), Bengali (bn), Finnish (fi), Japanese (ja), Korean (ko), Russian (ru) and Telugu (te):

Learning to rank model choice. We compare different learning to rank models as choices for learning the Bradley-Terry model coefficients. We conduct our experiments on the train set, where models contain pairwise judgments, and on a randomly sampled holdout set, a realistic scenario, with no available training data. We evaluate the following choices: Random Forest, Linear Regression, MLP Regressor, XGB Regressor, and SVR. All models are implemented via scikit-learn.⁸ The results are shown in Table 3. Random forest achieves the best R^2 metric on the holdout subset for 4 out of 7 languages. SVR also achieves a similar R^2 metric on the holdout subset, however, underperforms random forest on the training subset. Other baselines, such as XGB Regressor and MLP Regressor show signs of significant overfitting on the training subset, thereby underperforming random forest on the holdout subset.

⁸https://scikit-learn.org/stable/supervised_learning.html

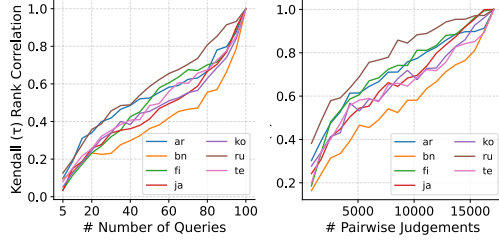


Figure 6: Sampling experiments to reduce computation cost. (left) reduces the number of queries whereas (right) reduces the pairwise judgments.

Features / Language	#F	ar	bn	fi	ja	ko	ru	te
(All) Features	11	0.938	0.867	0.921	0.881	0.921	0.826	0.918
(W/o) LLM as a Judge	9	0.912	0.866	0.898	0.853	0.891	0.811	0.904
(W/o) Low. Correlation	7	0.951	0.867	0.923	0.885	0.929	0.829	0.940
(Only) LLM as a Judge	2	0.948	0.728	0.907	0.884	0.916	0.851	0.872

Table 4: Kendall Tau (τ) scores using different features for training the random forest regression model.

Non-exhaustive pairwise comparisons lead to performance degradation. Exhaustive pairwise comparisons across a subset of 19 models in MIRAGE-BENCH using GPT-4o for all queries are quite expensive. To avoid this, we investigate whether all pairwise exhaustive comparisons are necessary during training. We utilize two sampling techniques: (i) full pairwise judgments on a sub-sample of 100 queries, e.g., 20 or 50 queries; (ii) partially judge a non-exhaustive random sample of the pairwise judgments across 100 queries, e.g., only 50% of all the exhaustive pairwise combinations. Both results are shown in Figure 6. We observe that Kendall-Tau (τ) correlations increase linearly with queries and pairwise judgments. In summary, an exhaustive pairwise comparison and a sufficient number of queries, such as 100, are necessary without impacting the leaderboard rankings.

All heuristic features are not necessary. We experiment with features used for random forest model training as a surrogate judge. We evaluate four training configurations: (i) all features (ii) without LLM-measured features (iii) without language detection and support, i.e., the low-correlation features observed in Figure 5, and (iv) including only LLM-measured features. From Table 4, we observe that removing low-correlated heuristic RAG features helps train the random forest model better leading to a conclusion that not necessarily all heuristic features are important. Removing the LLM-measured features completely or only using them for training the model decreases the Kendall-Tau (τ) correlation score.

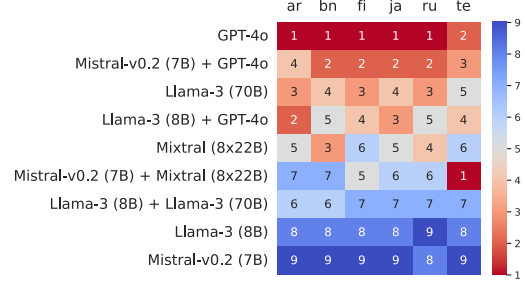


Figure 7: Approximate rankings using heuristic features after fine-tuning Llama-3 (8B) and Mistral-v0.2 (7B) on MIRAGE-BENCH dataset across four configurations.

Fine-tuning on MIRAGE-BENCH training data.

We evaluate three variants of the MIRAGE-BENCH training dataset using two smaller open-source LLMs: Mistral-v0.2 (7B) and Llama-3 (8B). We fine-tune the MIRAGE-BENCH training datasets using (i) both on GPT-4o, (ii) Llama-3 (8B) on Llama-3 (70B), and (iii) Mistral-v0.2 (7B) on Mistral (8×22B). From Figure 7, we observe that GPT-4o is a strong teacher, Mistral-v0.2 (7B) fine-tuned on GPT-4o distilled training data achieves rank 2 outperforming Llama-3 (70B). This shows that MIRAGE-BENCH training data is useful for improving the RAG answer generation task quality.

7 Conclusion

We present MIRAGE-BENCH, a multilingual RAG benchmark for 18 languages aimed at evaluating the multilingual generation part within RAG and aggregate traditional heuristic-based features to train a lightweight learning to rank model as a surrogate judge to learn a Bradley Terry model with GPT-4o pairwise judgments. Our results indicate a strong correlation between our surrogate judge and GPT-4o as a pairwise judge. This demonstrated the effectiveness of our efficient, cheap, and easy-to-retrain random forest model as a surrogate judge trained using only computationally cheap heuristic features for arena-based leaderboard ranking by achieving a 0.909 Kendall τ . On MIRAGE-BENCH, we observe that most proprietary and open-source larger LLMs currently dominate, whereas smaller open-source LLMs continue to struggle. Instruction tuning on MIRAGE-BENCH training data helps improve smaller open-source LLMs, e.g. instruction-tuned Mistral-v0.2 (7B) on GPT-4o distilled training data can outperform Llama 3 (70B) on MIRAGE-BENCH.

8 Limitations

MIRAGE-BENCH is one of the first large-scale multilingual RAG benchmarks. Although not perfect, we below discuss a set of limitations in our work:

- In MIRAGE-BENCH, we focused on benchmarking the generation task in RAG with oracle passages, we did not consider the retrieval task and its error propagating on the generation task.
- Due to budget constraints, we were unable to evaluate diverse LLMs as teachers such as Claude-3.5 (sonnet) (Anthropic, 2024) or Gemini Pro (Anil et al., 2023). We evaluated using GPT-4o which can cause self-enhancement bias towards LLMs in the OpenAI family.
- In our heuristic evaluation, we only considered a smaller subset of features. We did not explore more recent hand-crafted features such as nugget-based recall and precision (Pradeep et al., 2024b; Farzi and Dietz, 2024; Arabzadeh and Clarke, 2024; Lin and Demner-Fushman, 2005).
- Lastly, MIRAGE-BENCH does not provide human-labeled answers for queries across all languages and is limited to Wikipedia as the source.

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A Heuristic-based Evaluation: Features & Additional Details

1. Language Identification: In a multilingual RAG system, the output response should ideally be in the same language in which the user asked their query. To capture this feature, we attempt to identify which natural language the output response is in. We use `langid` (Lui and Baldwin, 2012), an off-the-shelf language detecting Python library for detecting the language of the long-form RAG answer. We use the probability of the target language detected as the score for language identification, i.e., $\hat{p} = \text{langid}(a, t)$, where t denotes the target language and a denotes the long-form answer.

2. Citation Quality: A multilingual RAG system must cite information from the relevant passages within their answers, to improve faithfulness and reduce hallucinations. We capture whether the passages (using relevance judgments provided in MIRACL (Zhang et al., 2023)) are cited in the multilingual generation task. For scoring, we compute the $\text{Recall}@k$ and $\text{MAP}@k$ score, where $\text{Recall}@k$ is 1.0 for a generated answer a , if and only if a cites all available relevant passages. Similarly, the $\text{MAP}@10$ score measures the percentage of relevant passages within the top- k cited passages.

3. Support: RAG systems have been shown to hallucinate across retrieval-augmented generation tasks, especially when provided with non-relevant contexts (Thakur et al., 2024b). Grounding is necessary to avoid hallucinations. We compute the grounding score of every sentence s_j in generated answer A along with the cited context c_j using the multilingual NLI model, which computes the similarity score as a probability of either *entailment*, *neutral* or *contradiction*. The entailment denotes the generated sentence in the long-form answer, which entails the cited passage within its response.

4. Reranker Score: The reranker score measures the semantic similarity between the user query and the cited passages in the system’s response. If the cited passages are relevant in answering the query, the reranker model would output a higher similarity score. We utilize a multilingual open-source reranker, namely BGE-M3 for our evaluation. We compute the reranker score across each cited passage p_i^j included in the long-form answer along with the user query q_i .

5. Answer Overlap: Existing open-domain question answering datasets such as Natural Questions

(Kwiatkowski et al., 2019), HotpotQA (Yang et al., 2018) or ELI5 (Fan et al., 2019) all include a human-labeled answer, assisting in evaluation using text overlap metrics such as Exact Match (EM) or F1. However, user queries in RAG systems potentially generate long-form answers, requiring metrics such as SacreBLEU or ROUGE-L to evaluate text-generation tasks. Automatic metrics are fairly quick and cheap to compute. For this reason, we include two metrics, SacreBLEU and ROUGE-L, for evaluating the RAG-generated answer. As we do not have human-labeled answers in MIRAGE-BENCH, we consider the GPT-4 generated answer as the gold truth for evaluation.

6. Answer overlap (LLM-measured): To capture semantic overlap between answers, we use the Llama-3 (8B) model as the judge for evaluation in a pointwise setup, where the LLM as a judge outputs a score between 1 to 5.

7. Fluency (LLM-measured): Fluency measures for grammatical correctness and idiomatic word choices in long-form answer generation. Evaluating fluency in multilingual long-form generation answers is not straightforward. While existing techniques are available for English such as MAUVE (Pillutla et al., 2021), only a few models evaluate multilingual summarization (Clark et al., 2023). Inspired by recent works in G-EVAL (Liu et al., 2023), we evaluate fluency using open-source LLM such as Llama-3 (8B) as the judge. Our reason for choosing open-source models lies in reducing the expense, of running an expensive proprietary LLM such as GPT-4. Our LLM as a judge setup outputs a score between 1 to 5.

B Baselines: Additional Details

In this section, we briefly describe each of the 19 multilingual-focused models utilized in our MIRAGE-BENCH evaluation experiments:

- 1. GPT-3.5-turbo:** (OpenAI, 2023) is evaluated using the Azure OpenAI service.⁹ We set the temperature parameter to 0.1 for a deterministic output. It utilizes the `cl100k_base` BPE-based tokenizer in the `tiktoken`¹⁰ repository.
- 2. GPT-4:** (OpenAI, 2023) is also evaluated using the Azure OpenAI service. We use a temperature setting of 0.1 for a deterministic output and the `cl100k_base` BPE-based tokenizer.

⁹<https://learn.microsoft.com/en-us/azure/ai-services/openai/>

¹⁰<https://github.com/openai/tiktoken>

3. **GPT-4o:** (OpenAI, 2023) is also evaluated using the Azure OpenAI service. We use a temperature setting of 0.1 for a deterministic output and the o200k_base BPE-based tokenizer.
4. **Mistral-7B-Instruct-v0.2** (Jiang et al., 2023) is the v0.2 of the instruct-version model containing 7B parameters.¹¹ It is an English-centric model, i.e., not instruction fine-tuned with any multilingual data. We used the multiple GPU inference using the vllm repository (Kwon et al., 2023). We set the temperature parameter to 0.1.
5. **Mistral-7B-Instruct-v0.3** (Jiang et al., 2023) is an extension of the instruct-version 0.2 model containing 7B parameters.¹² We set inference parameters similar to the previous model.
6. **Mixtral-8×7B-Instruct-v0.1** (Jiang et al., 2024) is a pretrained generative sparse Mixture of Experts (MoE), containing 8×7B parameters. It has been pretrained in 5 languages including English, French, Italian, German, and Spanish.¹³ As the model is computationally not feasible to evaluate due to resource constraints, We use the model API endpoint available in the Anyscale platform (<https://www.anyscale.com/>), with a temperature setting of 0.1.
7. **Mixtral-8×22B-Instruct-v0.1** (Jiang et al., 2024) is a pretrained generative sparse Mixture of Experts (MoE), containing 8×22B parameters. Similar to before, it has pretrained on 5 languages including English, French, Italian, German, and Spanish.¹⁴ We utilize the model API endpoint available in the Anyscale platform (<https://www.anyscale.com/>), with a temperature setting of 0.1.
8. **Command R** is developed keeping RAG in mind and officially supports 11 languages: Arabic, Brazilian, Portuguese, English, French, German, Italian, Japanese, Korean, Chinese, and Spanish. The model contains 35 billion parameters.¹⁵ We utilize the model API available in the Cohere platform (<https://cohere.com/>), with a temperature setting of 0.1, and using the chat template format.
9. **Command R+** is also developed keeping RAG in mind and officially supports 10 languages: English, French, Spanish, Italian, German, Brazilian Portuguese, Japanese, Korean, Arabic, and Chinese. The model contains 105 billion parameters.¹⁶ We utilize the model API available in the Cohere platform (<https://cohere.com/>), with a temperature setting of 0.1, and using the chat template format.
10. **Aya-23-35B** (Aryabumi et al., 2024) is an instruction fine-tuned model with highly advanced multilingual capabilities. The model officially supports 23 languages: Arabic, Chinese (simplified & traditional), Czech, Dutch, English, French, German, Greek, Hebrew, Hindi, Indonesian, Italian, Japanese, Korean, Persian, Polish, Portuguese, Romanian, Russian, Spanish, Turkish, Ukrainian, and Vietnamese. The model contains 35 billion parameters.¹⁷ We utilize the model API available in the Cohere platform (<https://cohere.com/>), with a temperature setting of 0.1, and using the chat template format.
11. **Gemma 1.1 (2B) it** (Mesnard et al., 2024) is an instruction fine-tuned model trained using the RLHF method containing 2 billion parameters.¹⁸ We used the multiple GPU inference using the vllm repository (Kwon et al., 2023). We set the temperature parameter to 0.1.
12. **Gemma 1.1 (7B) it** (Mesnard et al., 2024) is an instruction fine-tuned model trained using the RLHF method containing 7 billion parameters.¹⁹ We used the multiple GPU inference using the vllm repository (Kwon et al., 2023). We set the temperature parameter to 0.1.
13. **Meta-Llama-3-8B-Instruct** (Dubey et al., 2024) is an English-only instruction fine-tuned model containing 8 billion parameters.²⁰ We used the multiple GPU inference using the vllm repository (Kwon et al., 2023). We set the temperature parameter to 0.1.
14. **Meta-Llama-3-70B-Instruct** (Dubey et al., 2024) is an instruction fine-tuned model containing 70B parameters.²¹ As the model is computationally not feasible to evaluate due to resource constraints, We use the model API endpoint available in the Anyscale platform (<https://www.anyscale.com/>), with a temperature setting of 0.1.
15. **Phi-3 (mini)** (Abdin et al., 2024) is an English-focused instruction fine-tuned model trained

¹¹ mistralai/Mistral-7B-Instruct-v0.2

¹² mistralai/Mistral-7B-Instruct-v0.3

¹³ mistralai/Mixtral-8x7B-Instruct-v0.1

¹⁴ mistralai/Mixtral-8x22B-Instruct-v0.1

¹⁵ CohereForAI/c4ai-command-r-v01

¹⁶ CohereForAI/c4ai-command-r-plus

¹⁷ CohereForAI/aya-23-35B

¹⁸ google/gemma-1.1-2b-it

¹⁹ google/gemma-1.1-7b-it

²⁰ meta-llama/Meta-Llama-3-8B-Instruct

²¹ meta-llama/Meta-Llama-3-70B-Instruct

model containing 3.8 billion parameters.²² We used the multiple GPU inference using the vllm repository (Kwon et al., 2023). We set the temperature parameter to 0.1.

16. **Phi-3 (small)** (Abdin et al., 2024) is a multilingual instruction fine-tuned model trained model containing 8 billion parameters.²³ There is no available information on the number of languages covered by the model. We used the multiple GPU inference using the vllm repository (Kwon et al., 2023). We set the temperature parameter to 0.1.
17. **Phi-3 (medium)** (Abdin et al., 2024) is a multilingual instruction fine-tuned model trained model containing 14 billion parameters.²⁴ There is no available information on the number of languages covered by the model. We used the multiple GPU inference using the vllm repository (Kwon et al., 2023). We set the temperature parameter to 0.1.
18. **Qwen2-1.5B-Instruct** (Yang et al., 2024a) is an English-focused instruction fine-tuned model trained model containing 1.5 billion parameters.²⁵ We used the multiple GPU inference using the vllm repository (Kwon et al., 2023). We set the temperature parameter to 0.1.
19. **Qwen2-7B-Instruct** (Yang et al., 2024a) is an English-focused instruction fine-tuned model trained model containing 7 billion parameters.²⁶ We used the multiple GPU inference using the vllm repository (Kwon et al., 2023). We set the temperature parameter to 0.1.

C MIRAGE Fine-tuning Details

For multilingual RAG fine-tuning, we use teacher models to *distill* synthetic knowledge directly within smaller open-source models. We first generate RAG outputs on the MIRAGE-BENCH training dataset using three high-performing teacher models: (i) GPT-4o, (ii) Llama-3 (70B), and (iii) Mixtral (8×22B), and generate RAG output for queries in MIRAGE-BENCH training dataset using only relevant passages, i.e., without distracting the model with information from non-relevant passages. We filter out the teacher model responses and curate them to create the training dataset.

Next, using supervised fine-tuning (SFT) with

LoRA (Hu et al., 2022), we fine-tune two open-source models: (i) Llama-3 (8B) and (ii) Mistral-v0.2 (7B). Our hyperparameter choices are listed in Table 6. We use PEFT (Mangrulkar et al., 2022) and the alignment-handbook²⁷ (Tunstall et al., 2023) for supervised LoRA fine-tuning. We fine-tune four variants of models: (i) Mistral-v0.2 (7B) distilled using GPT-4o as a teacher, (ii) Mistral-v0.2 (7B) distilled using Mixtral (8×22B) and itself as a teacher, (iii) Llama-3 (8B) distilled using GPT-4o as a teacher, and (iv) Llama-3 (8B) distilled using Llama-3 (70B) and itself as a teacher. After fine-tuning, first all heuristic features are computed, using the already trained learning to rank model (using the baselines) is used to compute inference for the fine-tuned models and compared against upper-bound baselines, GPT-4o, Llama-3 (70B), and Mixtral (8×22B) and lower-bound baselines, Mistral-v0.2 (7B) and Llama-3 (8B).

D Extending MIRAGE Evaluation

As a holdout experiment, we evaluate newer versions of models, (i) Llama-3.1 series (Dubey et al., 2024): Llama-3.1 (8B)²⁸ and Llama-3.1 (70B)²⁹ instruct versions, and (ii) Gemma-2 series (Team et al., 2024): Gemma-2 (9B)³⁰ and Gemma-2 (27B)³¹ instruct versions. For both models, we used the API versions of the model provided by NVIDIA (<https://build.nvidia.com/>) by setting the temperature parameter to 0.1. The maximum sequence length of Gemma-2 models is 4096 tokens.

Experimental results. From Figure 8, we observe that the Gemma-2 (27B) and Llama-3.1 (70B) are strong baselines, by achieving an overall rank of 4 and 5 in the MIRAGE-BENCH dataset. Gemma-2 (27B) improves the previously best Gemma-1.1 (7B) by 13 ranks, whereas Llama-3.1 (70B) continues to underperform the best Llama-3 (70B) by 2 ranks. These results indicate newer models are improving, as reported using the surrogate judge on the synthetic MIRAGE-BENCH leaderboard.

²² [microsoft/Phi-3-mini-128k-instruct](https://github.com/microsoft/Phi-3-mini-128k-instruct)

²³ [microsoft/Phi-3-small-8k-instruct](https://github.com/microsoft/Phi-3-small-8k-instruct)

²⁴ [microsoft/Phi-3-medium-128k-instruct](https://github.com/microsoft/Phi-3-medium-128k-instruct)

²⁵ [Qwen/Qwen2-1.5B-Instruct](https://github.com/QwenLM/Qwen2-1.5B-Instruct)

²⁶ [Qwen/Qwen2-7B-Instruct](https://github.com/QwenLM/Qwen2-7B-Instruct)

²⁷ <https://github.com/huggingface/alignment-handbook>

²⁸ [meta-llama/Meta-Llama-3.1-8B-Instruct](https://huggingface.co/meta-llama/Meta-Llama-3.1-8B-Instruct)

²⁹ [meta-llama/Meta-Llama-3.1-70B-Instruct](https://huggingface.co/meta-llama/Meta-Llama-3.1-70B-Instruct)

³⁰ [google/gemma-2-9b-it](https://huggingface.co/google/gemma-2-9b-it)

³¹ [google/gemma-2-27b-it](https://huggingface.co/google/gemma-2-27b-it)

Lang.	Mean	95% CI	Lang.	Mean	95% CI	Lang.	Mean	95% CI
ar	0.916	-0.15 / +0.07	bn	0.937	-0.17 / +0.06	de	0.939	-0.14 / +0.05
en	0.971	-0.07 / +0.03	es	0.844	-0.12 / +0.09	fa	0.944	-0.22 / +0.05
fi	0.957	-0.07 / +0.04	fr	0.861	-0.15 / +0.09	hi	0.858	-0.26 / +0.13
id	0.793	-0.17 / +0.12	ja	0.892	-0.13 / +0.08	ko	0.941	-0.13 / +0.06
ru	0.968	-0.11 / +0.03	sw	0.973	-0.06 / +0.03	te	0.929	-0.16 / +0.07
th	0.902	-0.12 / +0.09	yo	0.709	-0.22 / +0.16	zh	0.954	-0.09 / +0.05

Table 5: $\bar{R}^2$ mean scores with 95% confidence interval with bootstrapping across all languages in MIRAGE-BENCH. We randomly kept two models as holdout in our work: Gemma 1.1 (2B) and Llama-3 (70B).

Hyper-parameter	Choice
Attention	FlashAttention-2 (Dao, 2024)
Batch Size	32
Epochs	1
Learning Rate	$2e-4$
Max Sequence Length	6144
Lora rank (r)	16
Lora alpha (α)	16
Lora dropout	0.05
Lora Modules	[q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj]

Table 6: Hyperparameter settings set during supervised fine-tuning of Mistral-v0.2 (7B) and Llama-3 (8B) on the MIRAGE-BENCH training dataset.

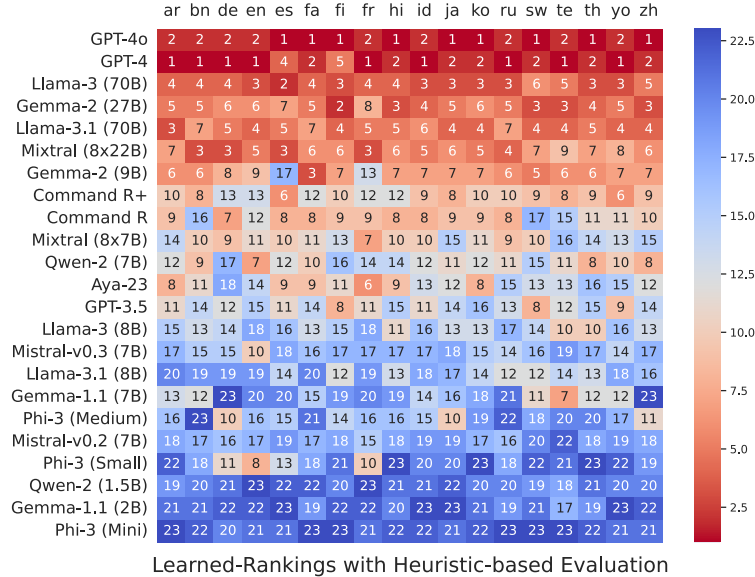


Figure 8: Approximate rankings using heuristic features including the newer models, Llama-3.1 (Dubey et al., 2024) and Gemma-2 (Team et al., 2024) on MIRAGE-BENCH dataset across all 18 languages. Gemma-2 (27B) and Llama-3.1 (70B) achieve a strong rank of 4 and 5 respectively in the MIRAGE-BENCH evaluation dataset.

1. The following table shows the number of persons employed in the principal occupations in the manufacturing industries of the State, by sex, race, and color, in 1900.

Occupation	Total		Male		Female	
	White	Colored	White	Colored	White	Colored
Manufacturing	1,234,567	234,567	1,000,000	200,000	234,567	34,567
Engineering	12,345	2,345	10,000	2,000	2,345	345
Machinery	234,567	45,678	200,000	40,000	34,567	5,678
Chemical	12,345	2,345	10,000	2,000	2,345	345
Textile	345,678	67,890	300,000	60,000	45,678	7,890
Food	456,789	89,012	400,000	80,000	56,789	9,012
Metals	567,890	112,345	500,000	100,000	67,890	12,345
Other	678,901	135,678	600,000	120,000	78,901	15,678

2. The following table shows the number of persons employed in the principal occupations in the manufacturing industries of the State, by sex, race, and color, in 1900.

Occupation	Total		Male		Female	
	White	Colored	White	Colored	White	Colored
Manufacturing	1,234,567	234,567	1,000,000	200,000	234,567	34,567
Engineering	12,345	2,345	10,000	2,000	2,345	345
Machinery	234,567	45,678	200,000	40,000	34,567	5,678
Chemical	12,345	2,345	10,000	2,000	2,345	345
Textile	345,678	67,890	300,000	60,000	45,678	7,890
Food	456,789	89,012	400,000	80,000	56,789	9,012
Metals	567,890	112,345	500,000	100,000	67,890	12,345
Other	678,901	135,678	600,000	120,000	78,901	15,678

3. The following table shows the number of persons employed in the principal occupations in the manufacturing industries of the State, by sex, race, and color, in 1900.

Occupation	Total		Male		Female	
	White	Colored	White	Colored	White	Colored
Manufacturing	1,234,567	234,567	1,000,000	200,000	234,567	34,567
Engineering	12,345	2,345	10,000	2,000	2,345	345
Machinery	234,567	45,678	200,000	40,000	34,567	5,678
Chemical	12,345	2,345	10,000	2,000	2,345	345
Textile	345,678	67,890	300,000	60,000	45,678	7,890
Food	456,789	89,012	400,000	80,000	56,789	9,012
Metals	567,890	112,345	500,000	100,000	67,890	12,345
Other	678,901	135,678	600,000	120,000	78,901	15,678

4. The following table shows the number of persons employed in the principal occupations in the manufacturing industries of the State, by sex, race, and color, in 1900.

Occupation	Total		Male		Female	
	White	Colored	White	Colored	White	Colored
Manufacturing	1,234,567	234,567	1,000,000	200,000	234,567	34,567
Engineering	12,345	2,345	10,000	2,000	2,345	345
Machinery	234,567	45,678	200,000	40,000	34,567	5,678
Chemical	12,345	2,345	10,000	2,000	2,345	345
Textile	345,678	67,890	300,000	60,000	45,678	7,890
Food	456,789	89,012	400,000	80,000	56,789	9,012
Metals	567,890	112,345	500,000	100,000	67,890	12,345
Other	678,901	135,678	600,000	120,000	78,901	15,678

5. The following table shows the number of persons employed in the principal occupations in the manufacturing industries of the State, by sex, race, and color, in 1900.

Occupation	Total		Male		Female	
	White	Colored	White	Colored	White	Colored
Manufacturing	1,234,567	234,567	1,000,000	200,000	234,567	34,567
Engineering	12,345	2,345	10,000	2,000	2,345	345
Machinery	234,567	45,678	200,000	40,000	34,567	5,678
Chemical	12,345	2,345	10,000	2,000	2,345	345
Textile	345,678	67,890	300,000	60,000	45,678	7,890
Food</						



Figure 9: Lollipop plots denoting the average heuristic-based feature scores achieved by baselines in MIRAGE-BENCH for all eleven heuristic-based features. x -axis denotes the languages in MIRAGE-BENCH; whereas y -axis plots every heuristic feature value. Multiple LLMs in the same family are represented as a single color lollipop (multiple circles).

Question:
What was the first newspaper ever printed in the U.K.?

Contexts:
 "[36897421#2]" Lögberg-Heimskringla - The very first newspaper to be published in North America by the Icelandic immigrant population was handwritten by Jon Gudmundsson in 1876 ...
 "[1965416#2]" The New York Times Magazine - Its first issue was published on September 6, 1896, and contained the first photographs ever printed in the newspaper...
 "[662134#6]" Letterpress printing - Letterpress printing was introduced in Canada in 1752 in Halifax, Nova Scotia by John Bushell in the newspaper format. This paper was named the Halifax Gazette and became Canada's first newspaper ...
 ...
 ...
 "[22112840#15]" Newspaper - The emergence of the new media in the 17th century has to be seen in close connection with the spread of the printing press from which the publishing press derives its name....

Instruction:
 Provide an answer to the question using the information provided in contexts written in {{language}}. Additionally, provide a step-by-step explanation of your reasoning, demonstrating how you arrived at your answer in {{language}}. Cite parts of your reasoning within brackets [] using the IEEE format based on the provided contexts.
 Please respond in {{language}} using the format: ##Reason: {reason} ##Answer: {answer}.

Figure 10: Prompt template for all baseline models for multilingual RAG generation for queries across all languages in MIRAGE-BENCH. We include the language-specific query in MIRAGE-BENCH under "Question:". Next, we concatenate both relevant and non-relevant passages (randomly shuffled and truncated at maximum length) and place them under "Contexts:". Lastly, we provide our instruction in English asking the model to generate a response in the required language under the placeholder "{{language}}". The example above is shown for a query in English (en) from MIRAGE-BENCH, where contexts are truncated (...) for demonstration purposes.

You are an AI assistant. In the following task, you are given a Question, a RAG application's response, and a Ground-truth Answer referred to as 'Label' in {{language}}. Assess how well the RAG application's response aligns with the Label, using the grading rubric below:

- 1: The response is not aligned with the Label or is off-topic; it includes hallucination.
- 2: The response admits it cannot provide an answer or lacks context; it is honest.
- 3: The response is relevant but contains notable discrepancies or inaccuracies.
- 4: The response is acceptable and sufficient but not exhaustive.
- 5: The response is fully accurate and comprehensive, based on the Label.

Treat the Label as the definitive answer. Present your final score in the format: "[[score]]", followed by your justification in English. Example:
 Score: [[3]] Justification: The response partially aligns with the Label but with some discrepancies.

Question in {{language}}:
 {{Question}}

Label in {{language}}:
 {{Label}}

RAG Application Response in {{language}}:
 {{Response}}

Treat the Label as the definitive answer. Present your final score in the format: "[[score]]", followed by your justification in English.

Figure 11: Prompt template used by Llama-3 (8B) model as a judge to evaluate the answer overlap heuristic feature. We include a grading rubric within the prompt template. {{Label}} is a placeholder for the gold truth answer provided using the GPT-4; {{language}} is a placeholder for the target language; {{Question}} is a placeholder for the MIRAGE-BENCH query; {{Documents}} is a placeholder for both MIRAGE-BENCH relevant and non-relevant passages concatenated together; {{Response}} is a placeholder for RAG model output.

You will be given one summary written for a question and documents from Wikipedia in {{language}}. Your task is to rate the summary on one metric. Please make sure you read and understand these instructions carefully. Please keep this document open while reviewing, and refer to it as needed.

Evaluation Criteria:
 Fluency (1-5) - the collective quality of all sentences. We align this dimension with the DUC quality question of structure and fluency whereby “the summary should be well-structured and well-organized. The summary should not just be a heap of related information, but should build from sentence to sentence to a coherent body of information about a topic.”

Evaluation Steps:
 1. Read the question and Wikipedia documents in {{language}} carefully and identify the main topic and key points.
 2. Read the summary and check whether it answers the question. Check if the summary covers the main topic and key points required to answer the question, and if it presents them in a clear and logical order.
 3. Assign a rating for fluency on a scale of 1 to 5 and provide an explanation, where 1 is the lowest and 5 is the highest based on the Evaluation Criteria.

Example:
 Question in {{language}}:
 {{Question}}

Documents in {{language}}:
 {{Documents}}

Summary:
 {{Summary}}

Rate the fluency of the summary on a scale of 1 to 5 and explain your rating. Please use the format of: ##Rating: {rating} ##Explanation: {explanation}.

Figure 12: Prompt template used by Llama-3 (8B) model as a judge to evaluate the fluency of a RAG response. We first explain the criteria for evaluation and the model outputs an explanation and score between [1,5] indicating the fluency of the output. {{language}} is a placeholder for the target language; {{Question}} is a placeholder for the MIRAGE-BENCH query; {{Documents}} is a placeholder for both MIRAGE-BENCH relevant and non-relevant passages concatenated together; {{Summary}} is a placeholder for RAG model output.

```

Please act as an impartial judge and evaluate the quality of the responses provided by two AI
assistants tasked to answer the question displayed below, based on a set of documents retrieved
by a search engine.
You should choose the assistant that best answers the user question based on a set of reference
documents that may or not be relevant referenced in the IEEE format.

Your evaluation should consider factors such as the correctness, helpfulness, completeness,
accuracy, depth, and level of detail of their responses.

Details are only useful if they answer the user's question. If an answer contains non-relevant
details, it should not be preferred over one that only uses relevant information.

Begin your evaluation by explaining why each answer correctly answers the user question.
Then, you should compare the two responses and provide a short explanation on their differences.
Avoid any position biases and ensure that the order in which the responses were presented does not
influence your decision. Do not allow the length of the responses to influence your evaluation.
Be as objective as possible.

After providing your explanation, output your final verdict by strictly following this
format: "[[A]]" if assistant A is better, "[[B]]" if assistant B is better, and "[[C]]" for a tie.

"[User Question]"
{{query}}

"[Reference Documents]"
{{documents}}

"[The Start of Assistant A's Answer]"
{{answer_a}}
"[The End of Assistant A's Answer]"

"[The Start of Assistant B's Answer]"
{{answer_b}}
"[The End of Assistant B's Answer]"

```

Figure 13: Prompt template used by LLM as a judge to evaluate the RAG response in a pairwise evaluation involving a head-to-head battle. The template is taken and modified from RAGEval (Rackauckas et al., 2024). We explain the evaluation criteria and ask the judge to evaluate two RAG responses based on multiple factors, including correctness, helpfulness, completeness, accuracy, depth, and level of detail. The Judge provides a justification for their model choice and at the end of the response indicates as either "[[A]]", "[[B]]", or "[[C]]" denoting a tie. `{{query}}` is a placeholder for the input MIRAGE-BENCH query; `{{documents}}` is a placeholder for both MIRAGE-BENCH relevant and non-relevant passages concatenated together; `{{answer_a}}` is a placeholder for the output response of model A; `{{answer_b}}` is a placeholder for the output response of model B.